



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
INNOVATION, SCIENCE AND TECHNOLOGY
DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

AUTOMOBILE ELECTRICS AND ELECTRONICS

**MODULE: Electrical and Electronics
Fundamentals**

PAPER NO: 306/22/M01

DURATION: 3 Hours

JUNE 2023 EXAMINATION

REQUIREMENTS

INSTRUCTIONS TO CANDIDATE

- 1. Section A: Answer ALL Questions (20 marks)**
- 2. Section B: Answer ALL Questions (20 marks)**
- 3. Section C: Answer Any four (4) Questions (60 marks)**

This paper consists of 6 printed pages.

Section A: Answer ALL Questions (20 marks)

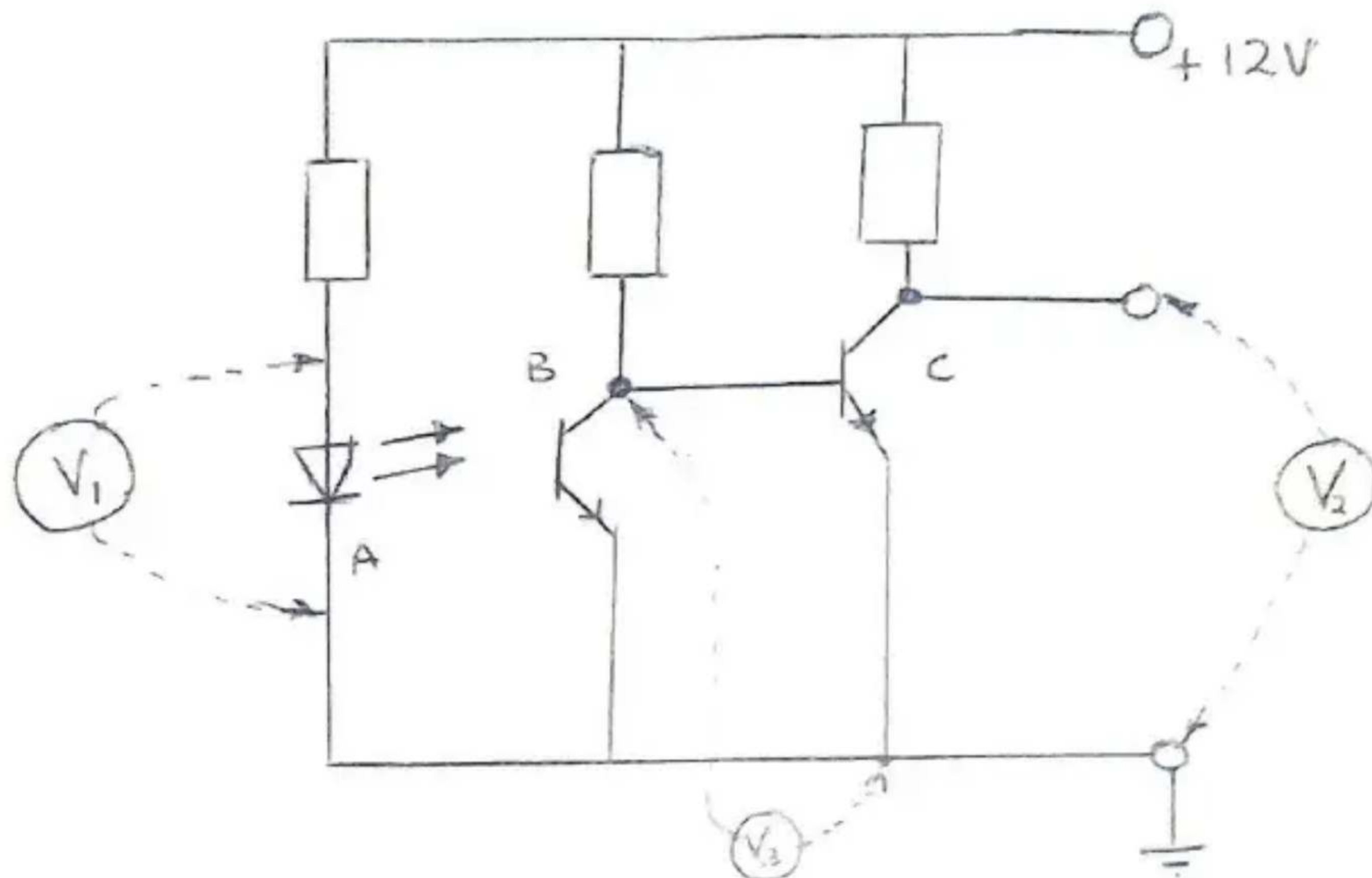
QUESTION 1

- a) Define the following electrical terms.
- | | | |
|------|--------------|----------|
| i) | Current | (1 mark) |
| ii) | Voltage | (1 mark) |
| iii) | Resistance | (1 mark) |
| iv) | Voltage drop | (1 mark) |
- b) State four (4) factors that influence voltage drop in a circuit. (4 marks)
- c) Identify any five (5) magnetic components on a motor vehicle. (5 marks)
- d) Two resistors of 3Ω and 6Ω are connected in parallel. Calculate the total resistance. (3 marks)
- e) Define eddy currents. (1 mark)
- f) State the meter that is used to measure the following electrical parameters.
- | | | |
|------|------------|----------|
| i) | Current | (1 mark) |
| ii) | Voltage | (1 mark) |
| iii) | Resistance | (1 mark) |

Section C: Answer Any four (4) Questions (60 marks)

QUESTION 3

Figure 1 shows an optoelectronic sensor circuit.

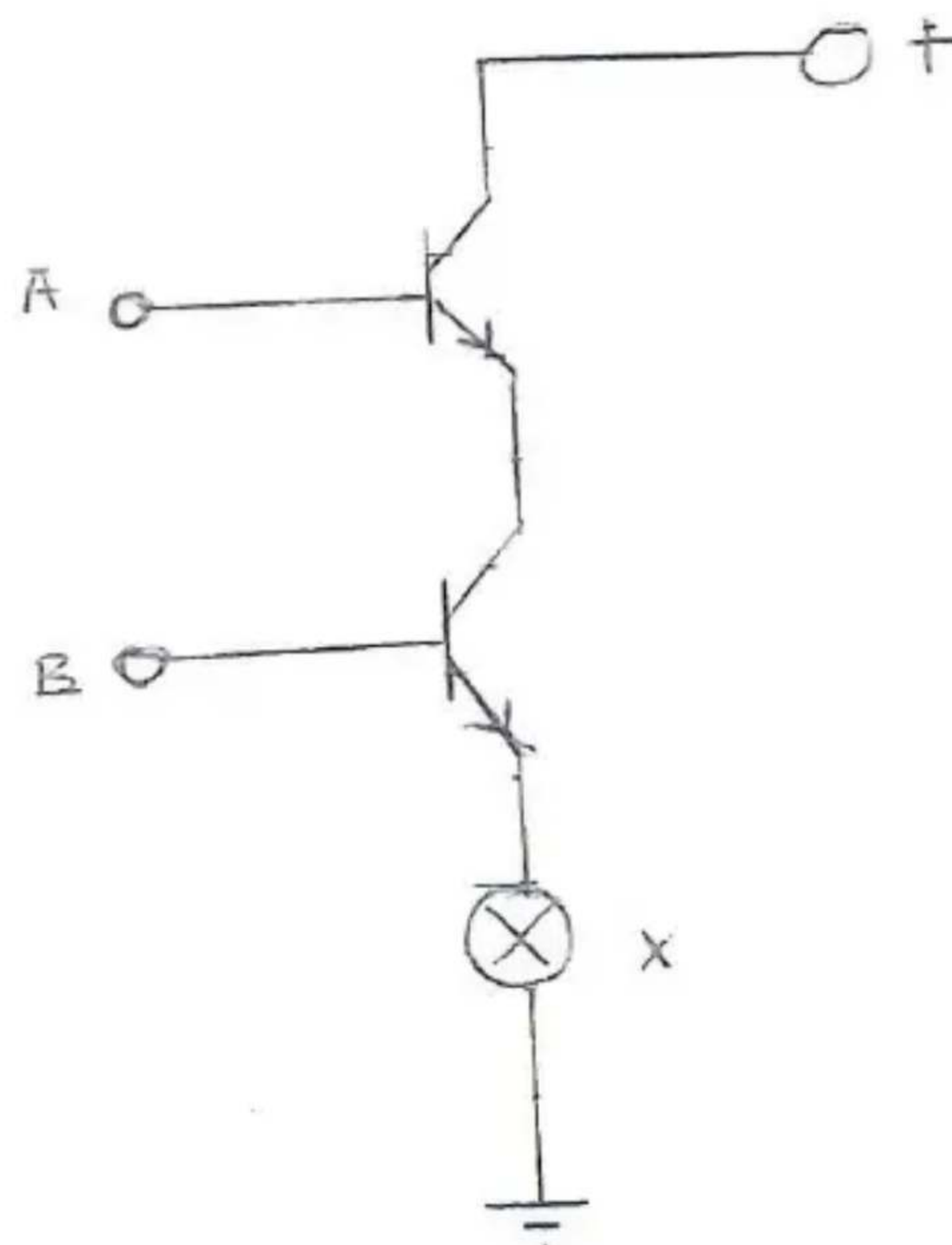


Study the diagram and answer the following questions.

- Name components A and B. (2 marks)
- Describe how A and B operate to turn C off. (4 marks)
- Reading on V_2 when item C is:
 - Switched on. (1 mark)
 - Switched off. (1 mark)
- State a typical reading for V_1 (2 marks)
 - Reading V_1 when item A is open circuited. (2 marks)
- If the voltmeter V_2 reads 12V and voltmeter V_3 reads 0.7V. State one fault that these reading indicates. (1 mark)
- Give two (2) application of opto-electronic sensors on a motor vehicle. (2 marks)

QUESTION 4

- a) Give three (3) basic logic gates. (3 marks)
- b) Logic circuits and systems formed on an IC use combination of
..... and (3 marks)
- c) Write the following abbreviations in full.
- | | | |
|------|------|----------|
| i) | TTL | (1 mark) |
| ii) | ECL | (1 mark) |
| iii) | CMOS | (1 mark) |
- d) Figure 2



- i) Name the logic gate represented by the transistor arrangement. (1 mark)
- i) Construct the truth table for the logic circuit. (4 marks)
- ii) State its boolean algebra. (1 mark)

QUESTION 5

- a) With the aid of a neat diagram explain the operation of a Wheatstone bridge (10 marks)
- b) List three (3) precautions that must be observed when using a multimeter. (3 marks)
- c) Name any two (2) multivibrator flip flop circuits. (2 marks)

QUESTION 6

- a) Identify four (4) basic functions of a vehicle computer. (4 marks)
- b) Outline three (3) variables that can be measured by sensors on the vehicle. (3 marks)
- c) Define the following:
 - i) Active sensor (1 mark)
 - ii) Actuator (1 mark)
- d) State three (3) major components of each of the following.
 - i) Hall sensor (3 marks)
 - ii) Magnetic variable reluctance (3 marks)

QUESTION 7

- a) Define the following terms:
 - i) Multiplexing (1 mark)
 - ii) Remote switching (1 mark)
- b) Explain the function of the following computer bus systems.
 - i) Data bus (1 mark)
 - ii) Address bus (2 marks)
 - iii) Control bus (2 marks)
- c) Indicate the eight step diagnostic procedure. (8 marks)